


# Aether Intelligent Messaging (AIM): Technical Introduction to AIM Wirelessware

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3Com
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# Introduction






- Why AIM?
- What Is AIM?
- How Do I Use AIM?
- What's Next?

# Challenges



- TCP Not Conducive to Wireless Communication
- Non-standard Communication Protocols
- Low throughput Rates
- Unfriendly Environment
- Multiple Networks
- Coverage Characteristics
- Enterprise Integration
- Mobile Device Selection and Network Dependencies

# What Is AIM?

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1. Optimized Wireless Messaging Transport
2. Software Development Environment

- Together they provide:
  - Enterprise connectivity
  - Mobile network independence
  - Mobile device independence
  - Rapid development

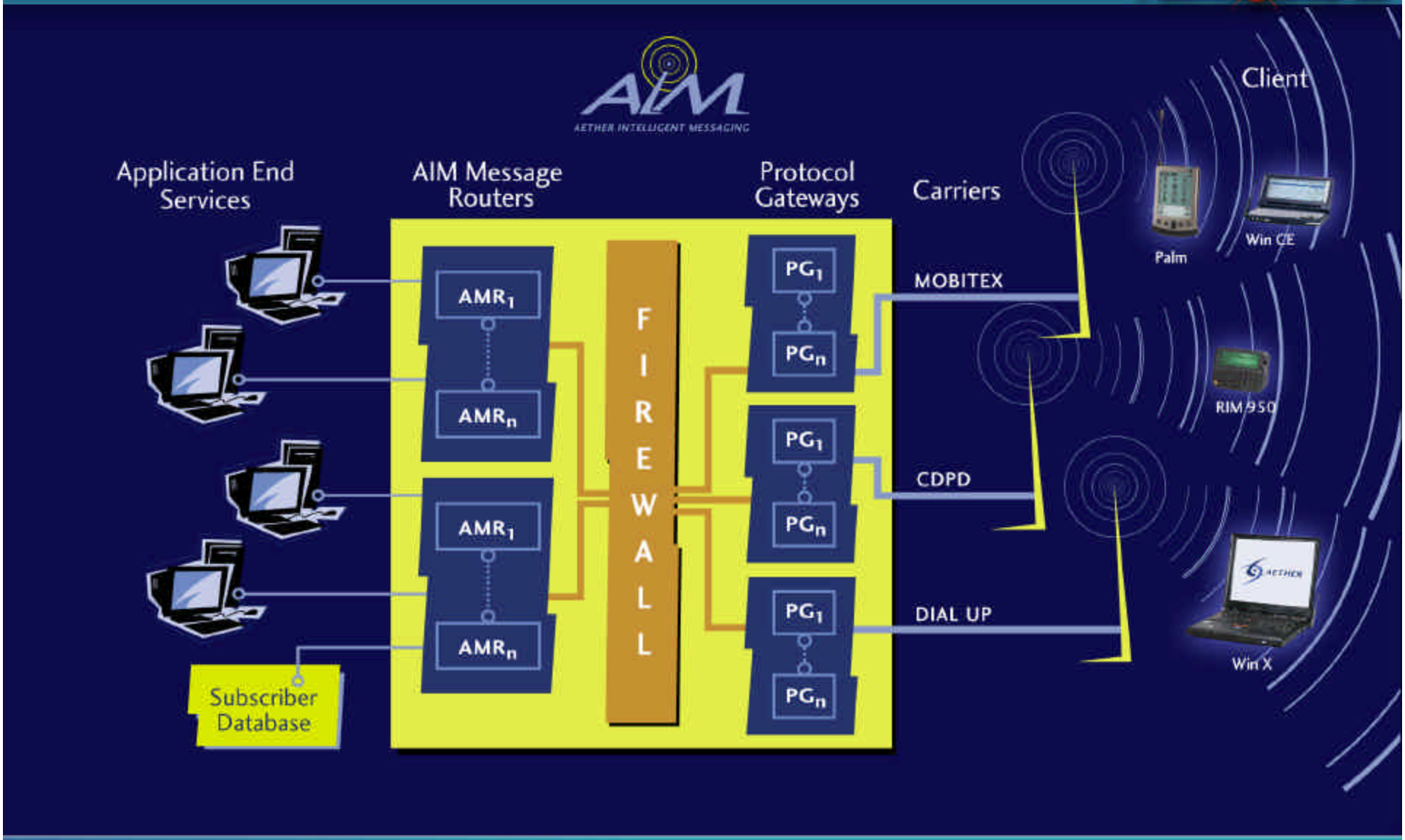
# AIM Benefits



- Optimized Airlink
- Compression / Security
- Network Independence
- Platform Independence
- Carrier-class Robustness
- Speed to Deployment

# Architectural Overview





# Airlink Optimization



- Standard Operation
  - Battery Management
  - Message oriented communication model
  - Confirmed delivery service
  - Retry management
  - Duplicate detection
- Segmentation and Reassembly
  - Segments large packets based on underlying network MTU size

# AIM vs. TCP: A Comparative Analysis

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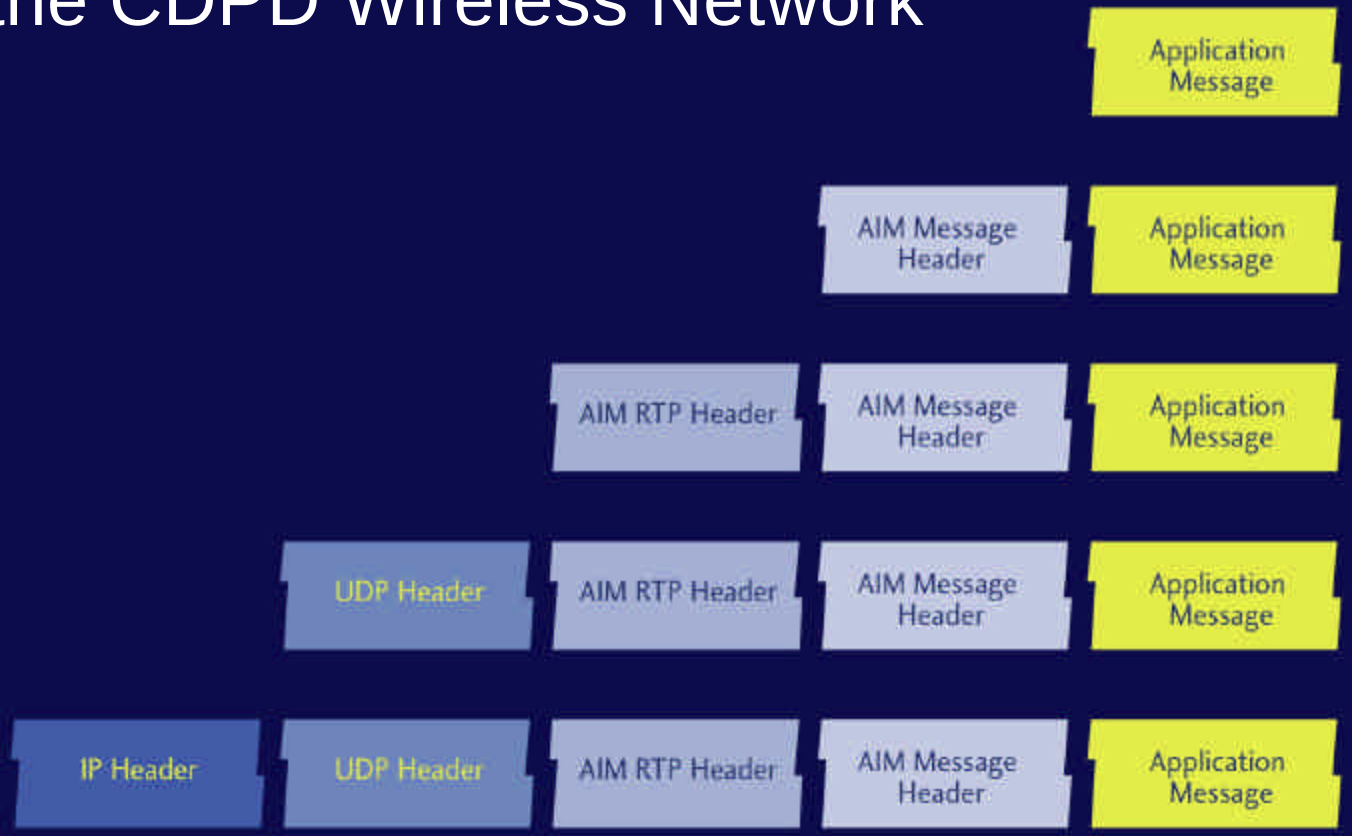
| | AIM | TCP | UDP |
|---|---|---|---|
| Reliable Delivery Service | yes | yes | no |
| Message Acknowledgement | yes | yes | no |
| Retransmission Management | yes | yes | no |
| Duplicate Detection | yes | yes | no |
| Segmentation/Reassembly | yes | yes | no |
| Connection Management Required | no | yes | no |

- Connection Oriented Semantics
- Connection-less Mode Generation
- Throughput Optimized

# AIM Message Overhead



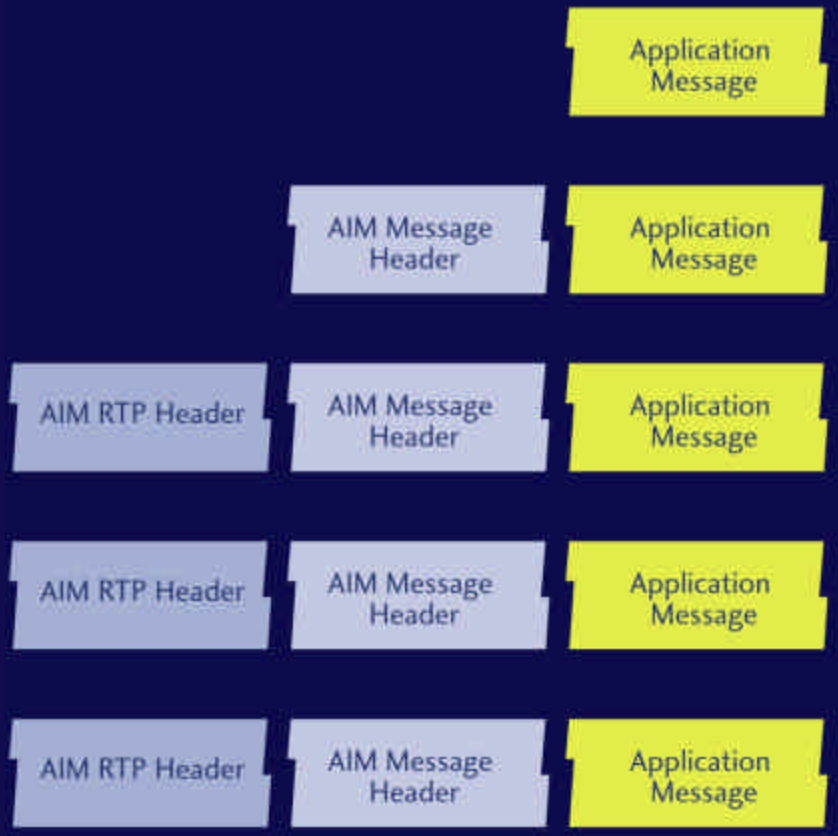
- Palm Computing® platform for the CDPD Wireless Network



# AIM Message Overhead



- Lightweight Reliable Transport
- 4B-6B Header
- Prepared by AIM Transport

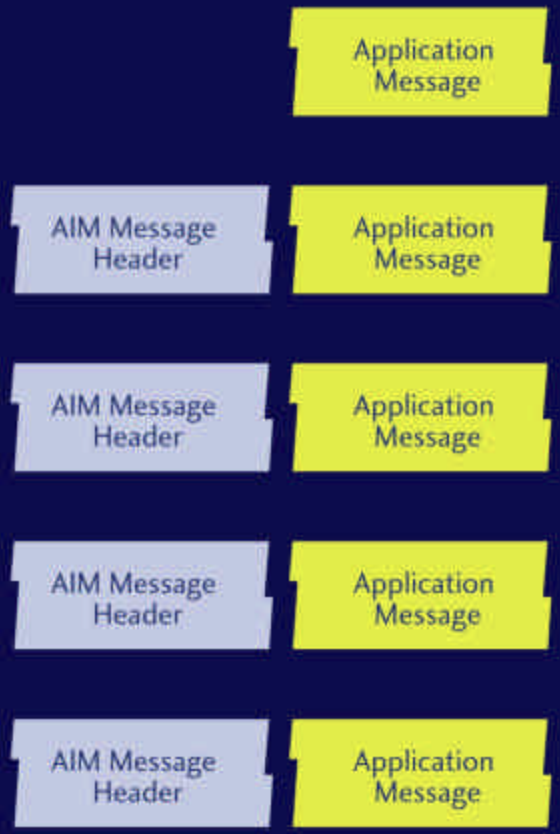


# AIM Message Overhead



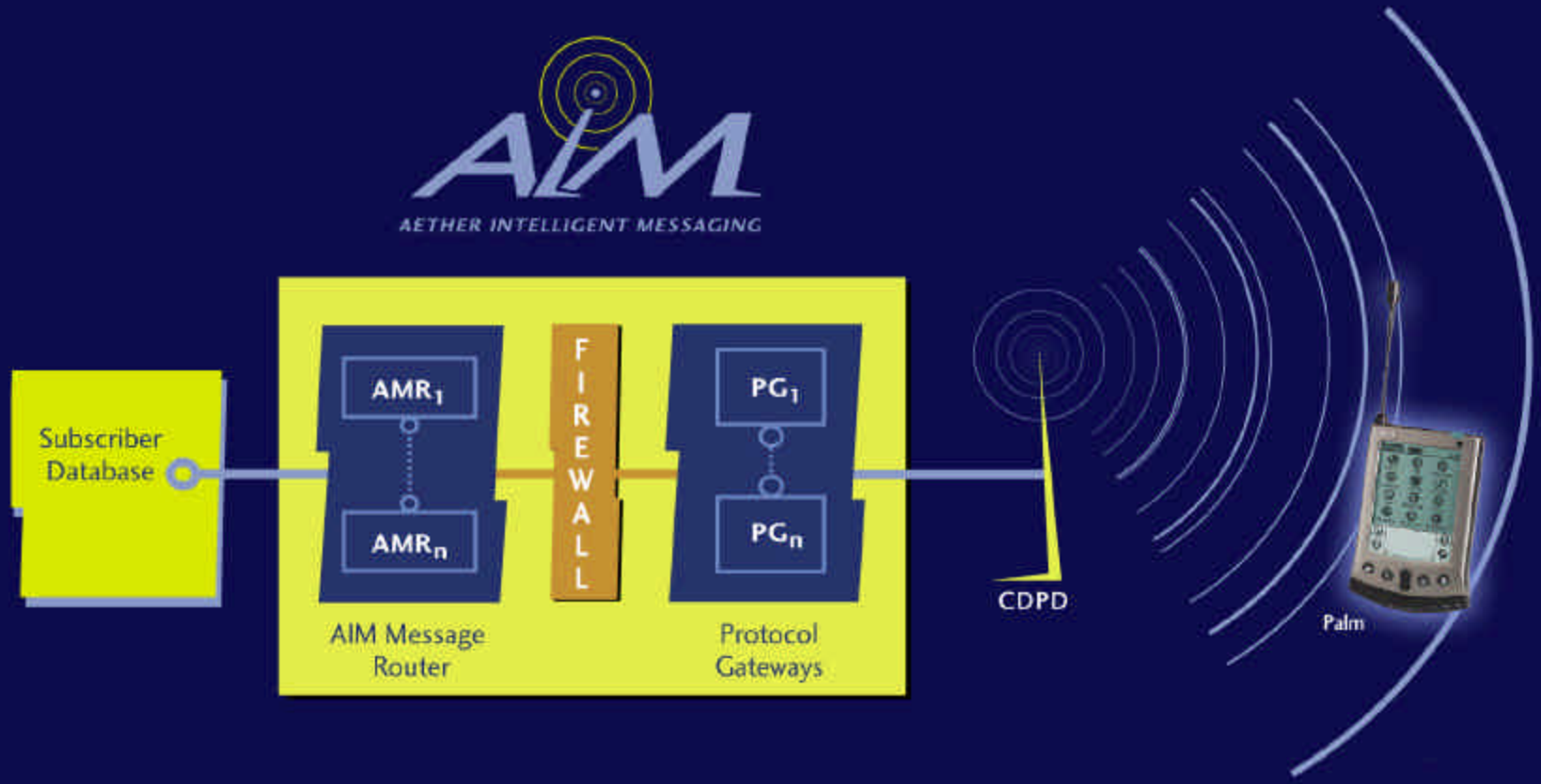
*AMR Uses AIM Msg Header to Route Messages*

- Service type
  - Quote service
- Server ID
  - Directs messages to a specific server for stateful communications
- Message ID
  - Uniquely identifies the message
- Message flags
  - AIM version info, compression, security



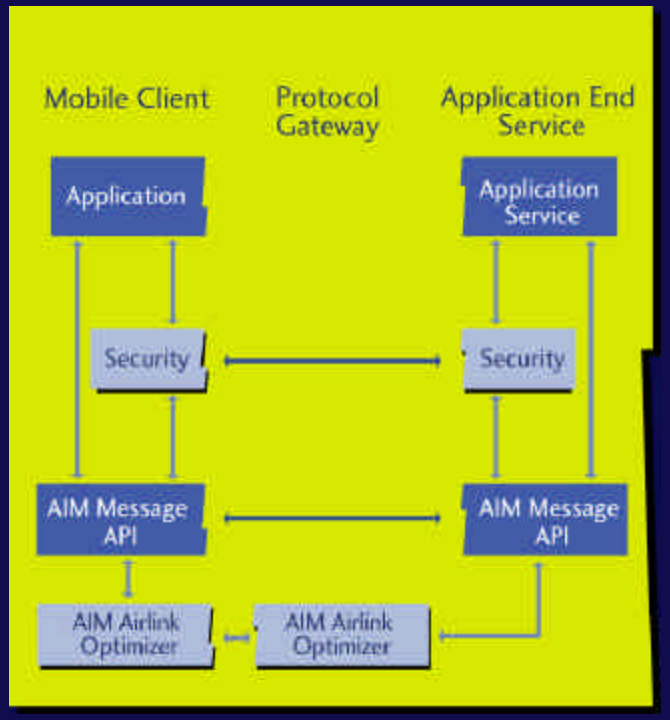
# Authenticated Level of Service





# Secure Communication Environment






- Elliptic Curve Cryptography – Licensed from Certicom
- End-to-End Security—Client to Service
- Equivalent to RSA 1024-bit
- Encryption
- Authentication
- Non-repudiation
- Message-based encryption not connection based

# Carrier Strength Implementation

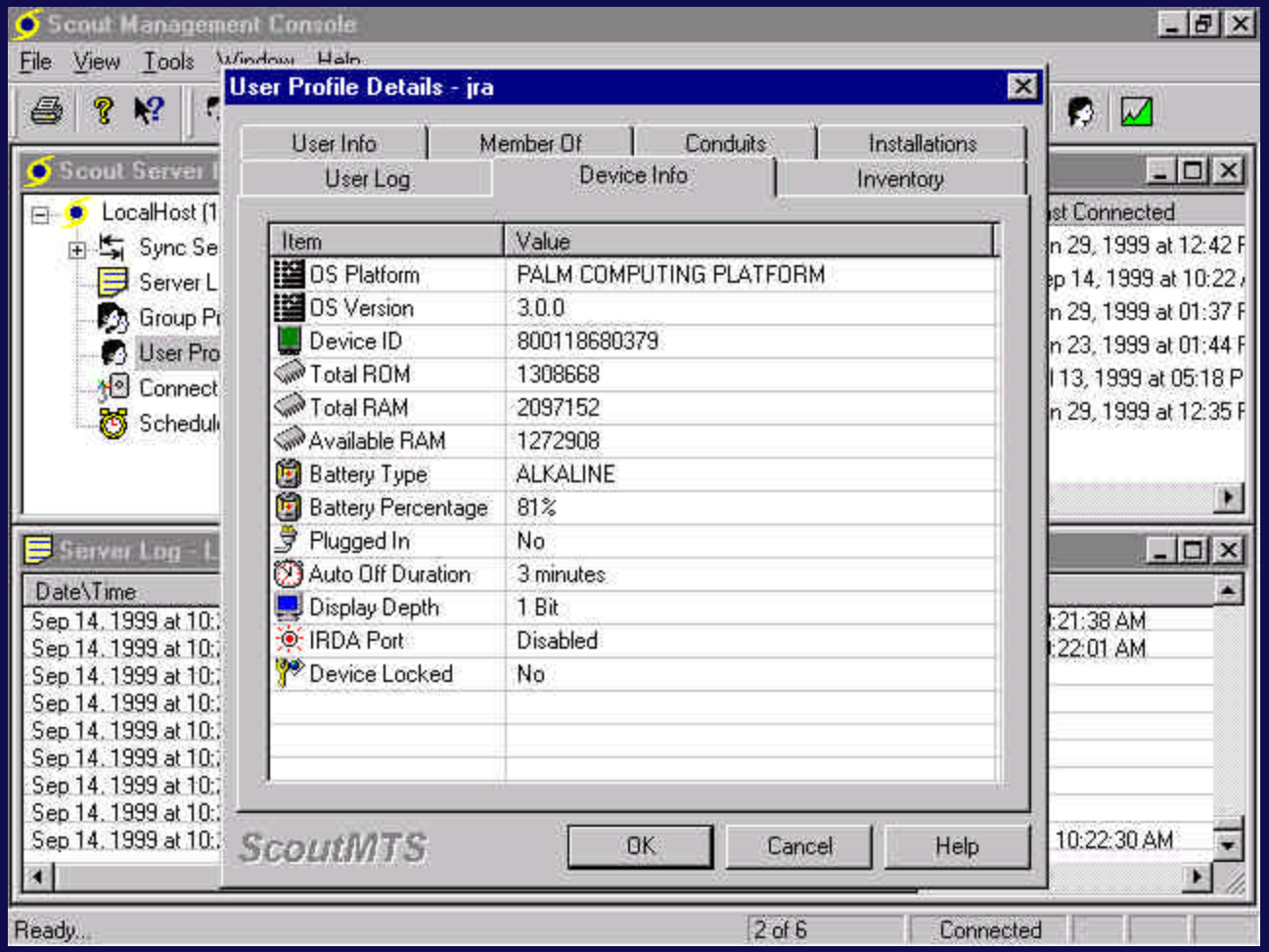


- Scalability and Load Balancing
- High Availability
- Manageable
- Transparent to the Client

# Aether/RiverBed Integration

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## ScoutIT—Managing the Mobile Enterprise




- Device Configuration
- Activity Tracking
- Individual and Group Monitoring
- Back-up and Restore
- Automated SUDS

# Application Services

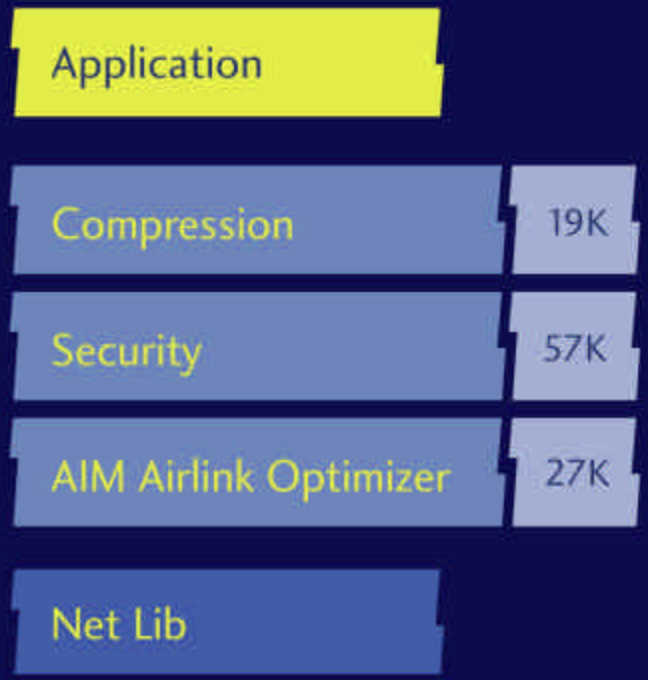





- Encapsulates Communication Interfaces
- Well Defined API Set
- Support for Secure Transaction Extensions
- Support for Application Data Compression

# Palm Implementation



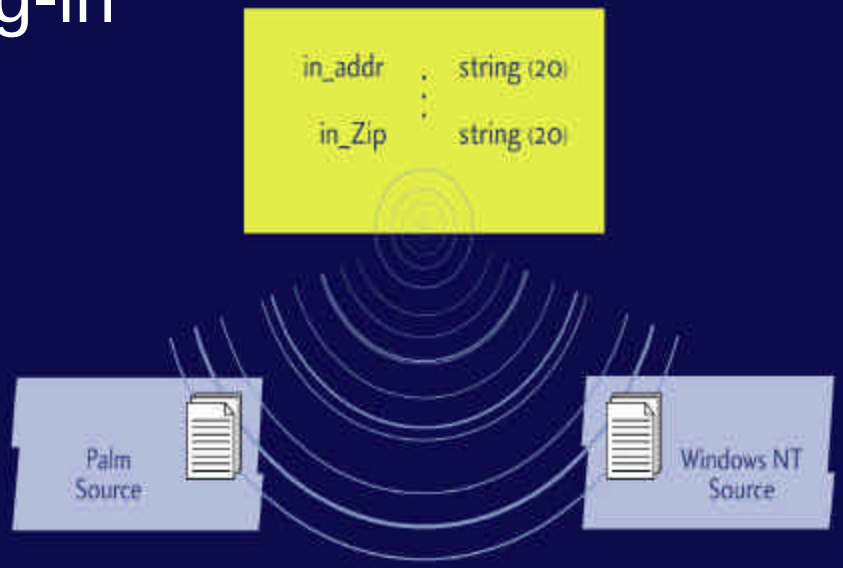



- Five (5) primary APIs
- Three (3) distinct shared software libraries

# AIM Message Wizard

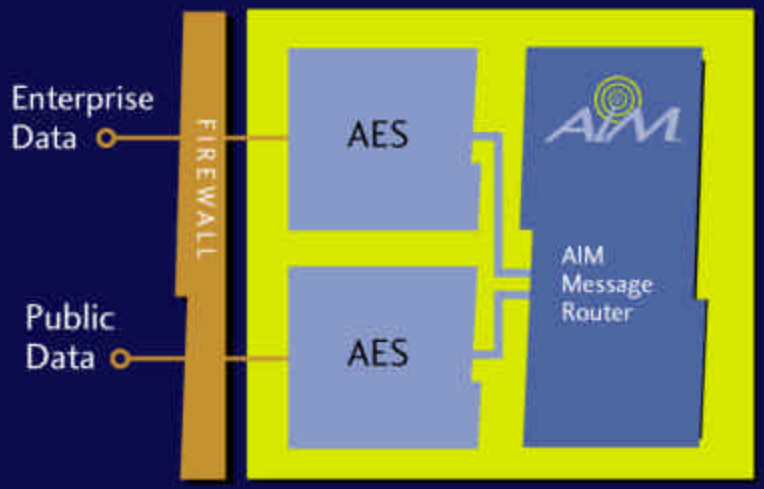


- Record-oriented Interface
- Automated Source Code Generation
- Implemented as Visual Studio Plug-in
- No Programming Required



# AIM AES Wizard





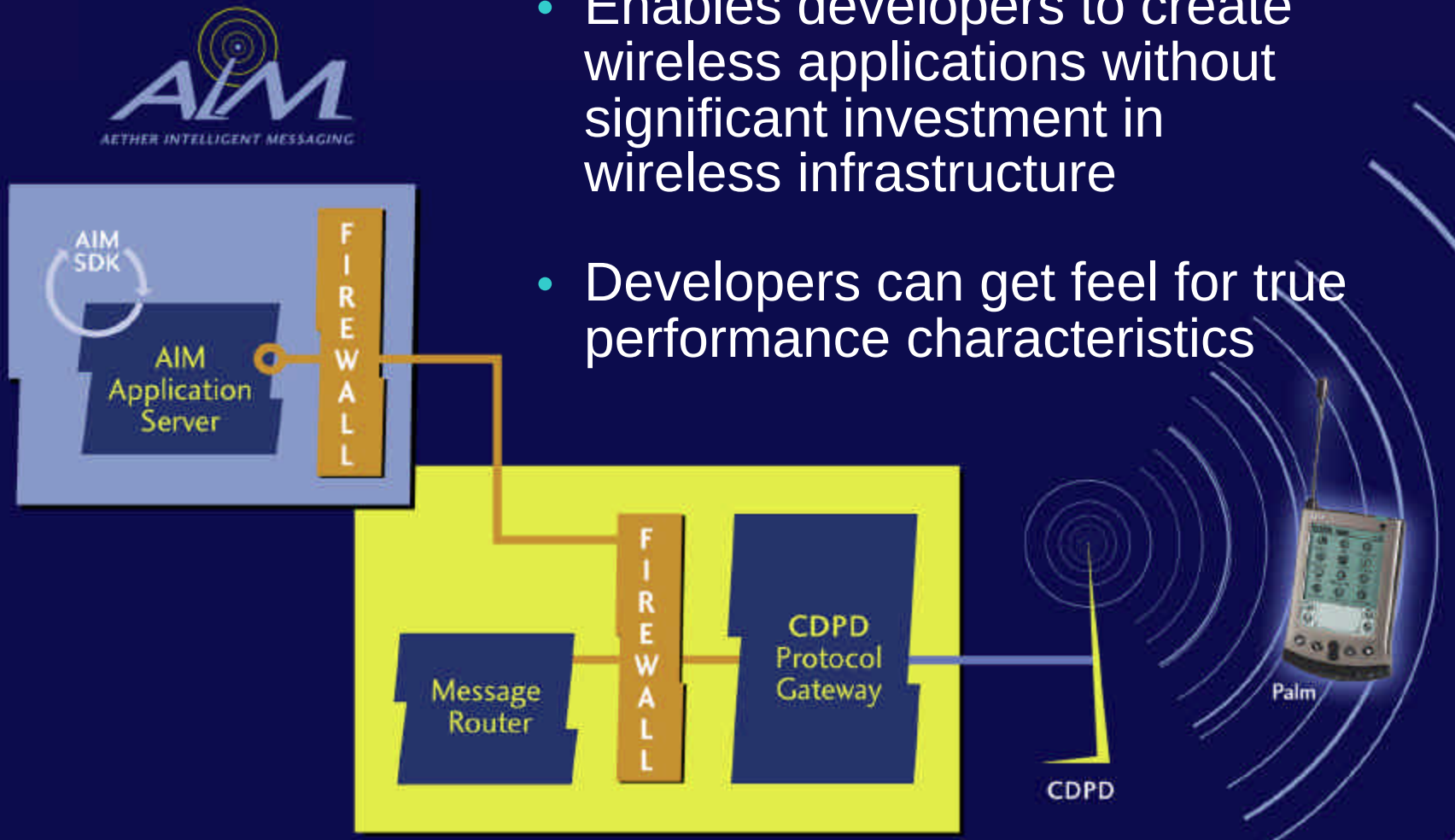

Source code generation for:

- AES communications with AIM infrastructure
- AES GUI
- AES Management functionality
- Implemented as Visual Studio plug-in

# AIM Developer's Network



- Enables developers to create wireless applications without significant investment in wireless infrastructure
- Developers can get feel for true performance characteristics



# Supported Platforms

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- Networks
  - CDPD
  - Mobitex
  - ISP Dial-up
  - 802.11
  - Wired LANs
- Platforms
  - Palm (3Com, Handspring, IBM, Qualcomm, Symbol)
  - CE 2.0/2.11 and Windows 95/98/NT
  - RIM 950 and Blackberry Pagers

# AIM Client SDK



- Multiple Platforms
- Multiple Networks
- Core Set of APIs
  - I/O Streaming Services
  - Compression Services
  - Transport Services
- Extended Set of APIs
  - Security Services

# What's Next?



- Partnerships
  - Riverbed
  - Oracle
- Other networks
  - Ricochet II
  - ReFlex
- Enterprise connectivity
  - Siebel
  - IBM MQ Series
- Customers
  - OpenSky
  - Merrill Lynch
  - Schwab
  - Discover
  - AAA
  - Others…

# Session #383, AIM Developer Lab

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- AIM Hands-On; Create a Wireless Palm Computing® Platform Application

  Friday, 10/21 at 12:30PM

  Location: Lab 1




- Agenda:
  - How do you use AIM?
  - Palm OS® SDK
  - Server SDK
  - Write a Palm Application

# Q&A

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Thank You!

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